

The image is a composite of three screenshots from a web browser, showing a Java programming lab interface. The browser's address bar shows the URL 'simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=692'. The page header features the 'SIMATS ENGINEERING' logo, the title 'SIMATS Online Programming Lab', the subtitle 'JAVA PROGRAMMING', and a user profile for 'YAMANURI PRANEETH' with ID '1923722698A'. The top screenshot displays a sidebar with five questions, the first of which is selected. The main content area shows a question titled 'Smart Document Editor using String, StringBuilder and StringBuffer' with 10 marks. The question lists seven tasks: 1. Read a paragraph from the user, 2. Count the number of characters, words, and sentences, 3. Replace every occurrence of a given word with another word, 4. Reverse each sentence individually, 5. Maintain editing history using StringBuffer, 6. Construct the final formatted document using StringBuffer, and 7. Display the original document, editing history, and final document. The requirement states to use String methods where appropriate, StringBuffer for document construction, and StringBuffer for edit history. The middle screenshot shows the 'Your Code' section with a Java program that implements these tasks. It uses Scanner for input, StringBuffer for history, and StringBuilder for document construction. The bottom screenshot shows a continuation of the Java code, focusing on the logic to reverse sentences and count characters, words, and sentences. The interface includes a taskbar at the bottom with various application icons and system status indicators like time (04:39 PM) and date (06-08-2026).

Q2

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=692`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subtitle "JAVA PROGRAMMING", and the user's name "YAMANURI PRANEETH" with a profile icon.

On the left sidebar, there are two questions listed:

- Q4: Digital Certificate Verification Using Set Interface (10 marks)
- Q5: Smart Inventory Management Using Map Interface (10 marks)

The main content area is titled "Requirement: Use a Suitable List Implementation". It contains a "Your Code" section with the following Java code:

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

public class HospitalTokenSystem {
    private static List<Patient> list = new ArrayList<>();

    public static void main(String[] args) {
        reg("P101", "Alice", "Cardio", 45);
        reg("P102", "Bob", "Neuro", 34);
        reg("P103", "Charlie", "Cardio", 60);
        showAll();
        update("P102", "Robert", "Neuro", 35);
        showDept("Cardio");
        cancel("P103");
        showAll();
        showFirstLast();
    }

    public static void reg(String id, String name, String dept, int age) {
        list.add(new Patient(id, name, dept, age));
        System.out.println("Registered. Token: " + list.size());
    }

    public static void showAll() {
        if (list.isEmpty()) return;
        for (int i = 0; i < list.size(); i++) {
            System.out.println("Token " + (i + 1) + ": " + list.get(i).toString());
        }
    }

    public static void update(String id, String name, String dept, int age) {
        Patient p = null;
        for (Patient patient : list) {
            if (patient.getId().equals(id)) {
                p = patient;
                break;
            }
        }
        if (p != null) {
            p.setName(name);
            p.setDept(dept);
            p.setAge(age);
        }
    }

    public static void cancel(String id) {
        list.removeIf(p -> p.getId().equalsIgnoreCase(id));
    }

    public static void showDept(String dept) {
        for (Patient p : list) {
            if (p.getDept().equalsIgnoreCase(dept)) System.out.println(p);
        }
    }

    public static void showFirstLast() {
        if (list.isEmpty()) {
            System.out.println("First: " + list.get(0));
            System.out.println("Last: " + list.get(list.size() - 1));
        }
    }
}

class Patient {
    String id, name, dept;
    int age;

    Patient(String id, String name, String dept, int age) {
        this.id = id;
        this.name = name;
        this.dept = dept;
        this.age = age;
    }

    public String toString() {
        return id + " | " + name + " | " + dept + " | " + age;
    }
}
```

The "Input" section shows the following data:

```
P101 Alice Cardio 45
P102 Bob Neuro 34
P103 Charlie Cardio 60
P102 Robert Neuro 35
```

The "Output" section shows the following results:

```
Registered. Token: 1
Registered. Token: 2
Registered. Token: 3
Token 1: P101 | Alice | Cardio | 45
Token 2: P102 | Bob | Neuro | 34
Token 3: P103 | Charlie | Cardio | 60
Token 4: P102 | Robert | Neuro | 35
Token 5: P101 | Alice | Cardio | 45
Token 6: P102 | Robert | Neuro | 35
Token 7: P101 | Alice | Cardio | 45
Token 8: P102 | Robert | Neuro | 35
Token 9: P101 | Alice | Cardio | 45
Token 10: P102 | Robert | Neuro | 35
Token 11: P101 | Alice | Cardio | 45
Token 12: P102 | Robert | Neuro | 35
Token 13: P101 | Alice | Cardio | 45
Token 14: P102 | Robert | Neuro | 35
Token 15: P101 | Alice | Cardio | 45
Token 16: P102 | Robert | Neuro | 35
Token 17: P101 | Alice | Cardio | 45
Token 18: P102 | Robert | Neuro | 35
Token 19: P101 | Alice | Cardio | 45
Token 20: P102 | Robert | Neuro | 35
Token 21: P101 | Alice | Cardio | 45
Token 22: P102 | Robert | Neuro | 35
Token 23: P101 | Alice | Cardio | 45
Token 24: P102 | Robert | Neuro | 35
Token 25: P101 | Alice | Cardio | 45
Token 26: P102 | Robert | Neuro | 35
Token 27: P101 | Alice | Cardio | 45
Token 28: P102 | Robert | Neuro | 35
Token 29: P101 | Alice | Cardio | 45
Token 30: P102 | Robert | Neuro | 35
Token 31: P101 | Alice | Cardio | 45
Token 32: P102 | Robert | Neuro | 35
Token 33: P101 | Alice | Cardio | 45
Token 34: P102 | Robert | Neuro | 35
Token 35: P101 | Alice | Cardio | 45
Token 36: P102 | Robert | Neuro | 35
Token 37: P101 | Alice | Cardio | 45
Token 38: P102 | Robert | Neuro | 35
Token 39: P101 | Alice | Cardio | 45
Token 40: P102 | Robert | Neuro | 35
Token 41: P101 | Alice | Cardio | 45
Token 42: P102 | Robert | Neuro | 35
Token 43: P101 | Alice | Cardio | 45
Token 44: P102 | Robert | Neuro | 35
Token 45: P101 | Alice | Cardio | 45
Token 46: P102 | Robert | Neuro | 35
Token 47: P101 | Alice | Cardio | 45
Token 48: P102 | Robert | Neuro | 35
Token 49: P101 | Alice | Cardio | 45
Token 50: P102 | Robert | Neuro | 35
Token 51: P101 | Alice | Cardio | 45
Token 52: P102 | Robert | Neuro | 35
Token 53: P101 | Alice | Cardio | 45
Token 54: P102 | Robert | Neuro | 35
Token 55: P101 | Alice | Cardio | 45
Token 56: P102 | Robert | Neuro | 35
Token 57: P101 | Alice | Cardio | 45
Token 58: P102 | Robert | Neuro | 35
Token 59: P101 | Alice | Cardio | 45
Token 60: P102 | Robert | Neuro | 35
Token 61: P101 | Alice | Cardio | 45
Token 62: P102 | Robert | Neuro | 35
Token 63: P101 | Alice | Cardio | 45
Token 64: P102 | Robert | Neuro | 35
Token 65: P101 | Alice | Cardio | 45
Token 66: P102 | Robert | Neuro | 35
Token 67: P101 | Alice | Cardio | 45
Token 68: P102 | Robert | Neuro | 35
Token 69: P101 | Alice | Cardio | 45
Token 70: P102 | Robert | Neuro | 35
Token 71: P101 | Alice | Cardio | 45
Token 72: P102 | Robert | Neuro | 35
Token 73: P101 | Alice | Cardio | 45
Token 74: P102 | Robert | Neuro | 35
Token 75: P101 | Alice | Cardio | 45
Token 76: P102 | Robert | Neuro | 35
Token 77: P101 | Alice | Cardio | 45
Token 78: P102 | Robert | Neuro | 35
Token 79: P101 | Alice | Cardio | 45
Token 80: P102 | Robert | Neuro | 35
Token 81: P101 | Alice | Cardio | 45
Token 82: P102 | Robert | Neuro | 35
Token 83: P101 | Alice | Cardio | 45
Token 84: P102 | Robert | Neuro | 35
Token 85: P101 | Alice | Cardio | 45
Token 86: P102 | Robert | Neuro | 35
Token 87: P101 | Alice | Cardio | 45
Token 88: P102 | Robert | Neuro | 35
Token 89: P101 | Alice | Cardio | 45
Token 90: P102 | Robert | Neuro | 35
Token 91: P101 | Alice | Cardio | 45
Token 92: P102 | Robert | Neuro | 35
Token 93: P101 | Alice | Cardio | 45
Token 94: P102 | Robert | Neuro | 35
Token 95: P101 | Alice | Cardio | 45
Token 96: P102 | Robert | Neuro | 35
Token 97: P101 | Alice | Cardio | 45
Token 98: P102 | Robert | Neuro | 35
Token 99: P101 | Alice | Cardio | 45
Token 100: P102 | Robert | Neuro | 35
```

The bottom of the screenshot shows the Windows taskbar with various application icons and the system clock indicating 04:42 PM on 06-08-2026.

Q3

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=692". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "YAMANURI PRANEETH" with a dropdown arrow. The left sidebar contains a list of questions, with the first two marked as answered. The main area is divided into three sections: "Your Code", "Input", and "Output".

Your Code:

```
import java.util.*;

public class AirportBoarding {
    public static void main(String[] args) {

        Queue<String> boardingQueue = new LinkedList<>();

        boardingQueue.add("Rahul");
        boardingQueue.add("Priya");
        boardingQueue.add("Kiran");
        boardingQueue.add("Anjali");

        System.out.println("Standard Boarding Queue: " + boardingQueue);

        System.out.println("Next Passenger: " + boardingQueue.peek());

        System.out.println("Boarded Passenger: " + boardingQueue.poll());

        System.out.println("Remaining Queue: " + boardingQueue);

        PriorityQueue<Integer> vipQueue = new PriorityQueue<>();

        vipQueue.add(1);
        vipQueue.add(3);
        vipQueue.add(2);

        System.out.println("VIP Priorities: " + vipQueue);

        while (!vipQueue.isEmpty()) {
            System.out.println("VIP Boarded (Priority: " + vipQueue.poll() + ")");
            System.out.println("Remaining VIP Queue: " + vipQueue);
        }
    }
}
```

Input:

Enter number of standard passengers:
4

Output:

```
Standard Boarding Queue: [Rahul, Priya, Kiran, Anjali]
Next Passenger: Rahul
Boarded Passenger: Rahul
Remaining Queue: [Priya, Kiran, Anjali]
VIP Priorities: [1, 3, 2]
VIP Boarded (Priority: 1)
Remaining VIP Queue: [3, 2]
VIP Boarded (Priority: 3)
Remaining VIP Queue: [2]
VIP Boarded (Priority: 2)
Remaining VIP Queue: []
```

Q4

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=692". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "YAMANURI PRANEETH" with a dropdown arrow. The left sidebar contains a list of questions, with the first two marked as answered. The main area is divided into three sections: "Your Code", "Input", and "Output".

Your Code:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        LinkedHashSet<String> certificates = new LinkedHashSet<>();

        certificates.add("CERT101");
        certificates.add("CERT102");
        certificates.add("CERT103");
        certificates.add("CERT104");
        certificates.add("CERT102");

        System.out.println("Certificates (Insertion Order):");
        System.out.println(certificates);

        String search = "CERT103";
        if (certificates.contains(search)) {
            System.out.println(search + " Found");
        } else {
            System.out.println(search + " Not Found");
        }

        TreeSet<String> sorted = new TreeSet<>(certificates);
        System.out.println("Certificates (Sorted Order):");
        System.out.println(sorted);

        System.out.println("Total Unique Certificates: " + certificates.size());
        certificates.remove("CERT104");
        System.out.println("After Removing Revoked Certificate:");
        System.out.println(certificates);
    }
}
```

Input:

No Input Required

Output:

```
Certificates (Insertion Order):
[CERT101, CERT102, CERT103, CERT104]
CERT103 Found
Certificates (Sorted Order):
[CERT101, CERT102, CERT103]
Total Unique Certificates: 4
After Removing Revoked Certificate:
[CERT101, CERT102, CERT103]
```

Q5

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=692`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "YAMANURI PRANEETH" with a profile icon.

The main workspace is divided into three sections:

- Left Panel:** A small notification box says "Using Web Interface (10 marks)" and another says "5/5 answered".
- Center Panel:** A code editor with a dark background showing a Java program. The code defines a `Product` class with attributes `name`, `category`, `quantity`, and `price`. It includes a constructor and a `main` method that uses a `HashMap` to store product data. The `main` method iterates through the map, prints product details, and calculates the total inventory value.
- Right Panel:** An "Output" window showing the execution results of the program.

The Java code in the center panel is as follows:

```
1 public class Product {
2     String name, category;
3     int quantity;
4     double price;
5
6     Product(String name, String category, int quantity, double price) {
7         this.name = name;
8         this.category = category;
9         this.quantity = quantity;
10        this.price = price;
11    }
12
13    public static void main(String[] args) {
14
15        HashMap<Integer, Product> map = new HashMap<>();
16
17        map.put(101, new Product("Laptop", "Electronics", 5, 5000));
18        map.put(102, new Product("Mouse", "Electronics", 10, 800));
19        map.put(103, new Product("Book", "Education", 20, 300));
20
21        map.get(103).quantity = 15;
22
23        int id = 102;
24
25        if (map.containsKey(id))
26            System.out.println("Product Found: " + map.get(id).name);
27        else
28            System.out.println("Product Not Found");
29
30        double total = 0;
31
32        for (Product p : map.values())
33            total += p.quantity * p.price;
34
35        System.out.println("Total Inventory Value: " + total);
36
37        System.out.println("Category-Wise Products:");
38
39        for (Map.Entry<Integer, Product> e : map.entrySet()) {
40            Product p = e.getValue();
41            System.out.println(p.category + " : " + p.name);
42        }
43
44        map.remove(103);
45    }
46 }
```

The output window on the right shows the following results:

```
Product Found: Mouse
Total Inventory Value: 26500.0
Category-Wise Products:
Electronics : Laptop
Electronics : Mouse
Education : Book
```

The Windows taskbar at the bottom shows the system clock as 04:46 PM on 06-08-2026, with a battery level of 88% and a weather forecast of "Mostly cloudy".